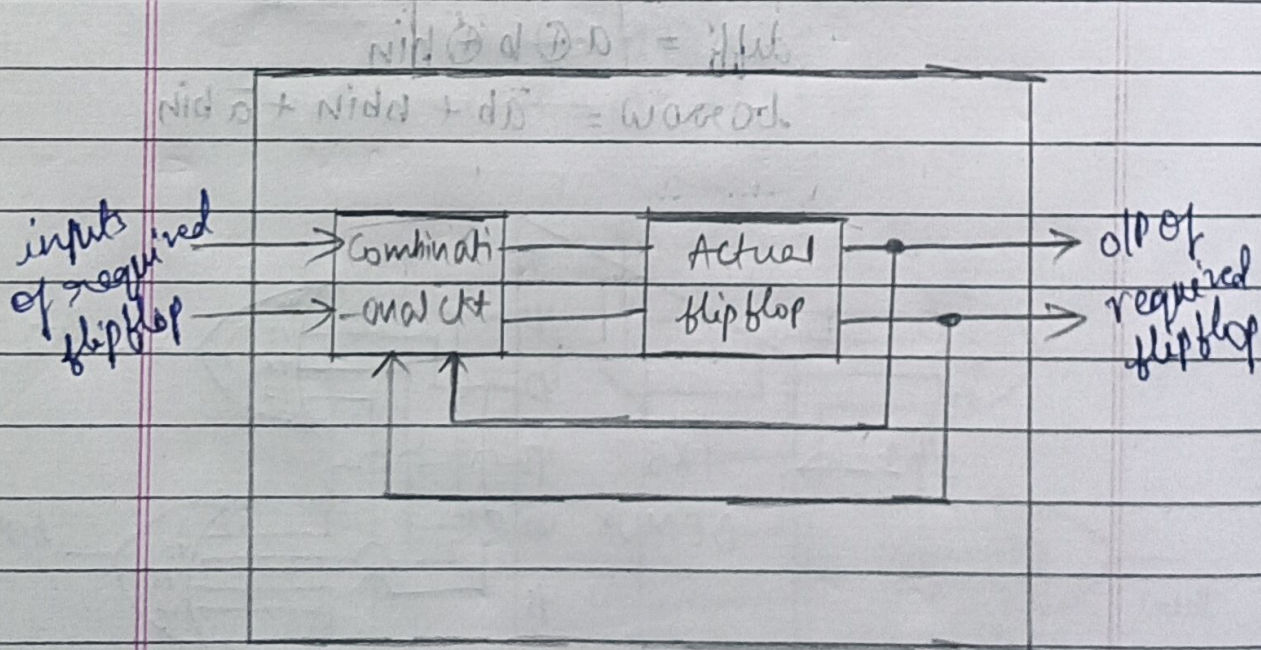


FLIP-FLOP CONVERSIONS

→ Converting one flip-flop into another flip-flop by adding some additional logic to the available flip-flop.

→ Steps for flipflop conversion :-

- ① Determine the required flipflop
- ② Draw the truth table of required flipflop.
- ③ With the help of excitation table of the available flipflop, determine the inputs to be applied for obtaining the required output.
- ④ Get the expressions of the input using K-Maps.
- ⑤ Implement the circuit by connecting the addition logic to the available flipflop.



1. SR flipflop to other flipflops

① SR to JK flipflop

→ The required flipflop is JK flipflop.

→

J	K	Q_n	Q_{n+1}	S	R
0	0	0	0	0	X
0	0	1	1	X	0
0	1	0	0	0	X
0	1	1	0	0	1
1	0	0	1	1	0
1	0	1	1	X	0
1	1	0	1	1	0
1	1	1	0	0	1

→ K-Maps

S:-

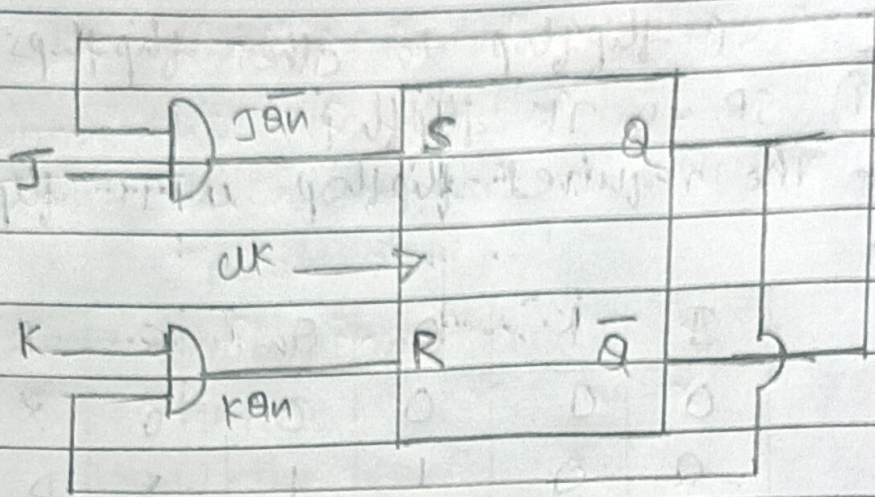
J \ K Q_n	00	01	11	10
0	1	X	1	0
1	1	X	1	1

$S = J \bar{Q}_n$

R:-

J \ K Q_n	00	01	11	10
0	X		1	X
1			1	

$R = K Q_n$



SR to JK flip flop

(2) SR to D flip flop

→ The required flip flop is D flip flop

→

	D	Q _n	Q _{n+1}	S	R
	0	0	0	0	X
	0	1	0	0	1
	1	0	1	1	0
	1	1	1	X	0

→ K-Maps

SR

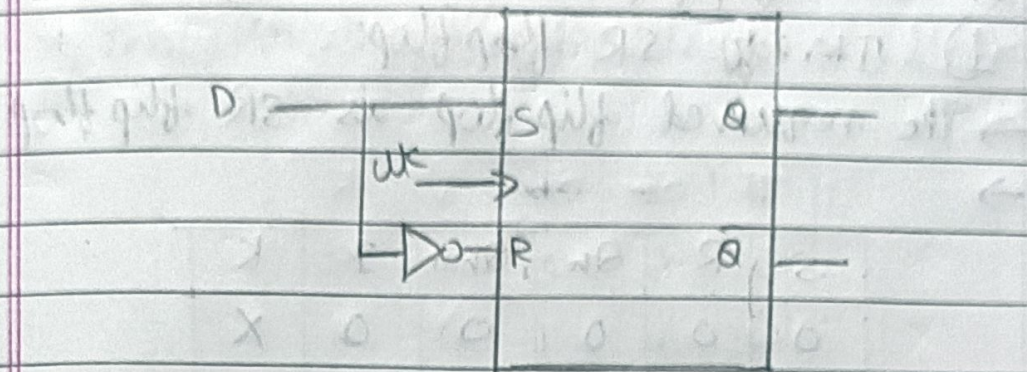
	Q _n	0	1
S	0		
1	1	1	X

$S = D$

R

	Q _n	0	1
R	0	X	1
1			

$R = \bar{D}$



SR to D flip-flop

③ SR to T flip-flop

→ the required flip-flop is T-flip-flop

→

T	Q_n	Q_{n+1}	S	R
0	0	0	0	X
0	1	1	X	0
1	0	1	1	0
1	1	0	0	1

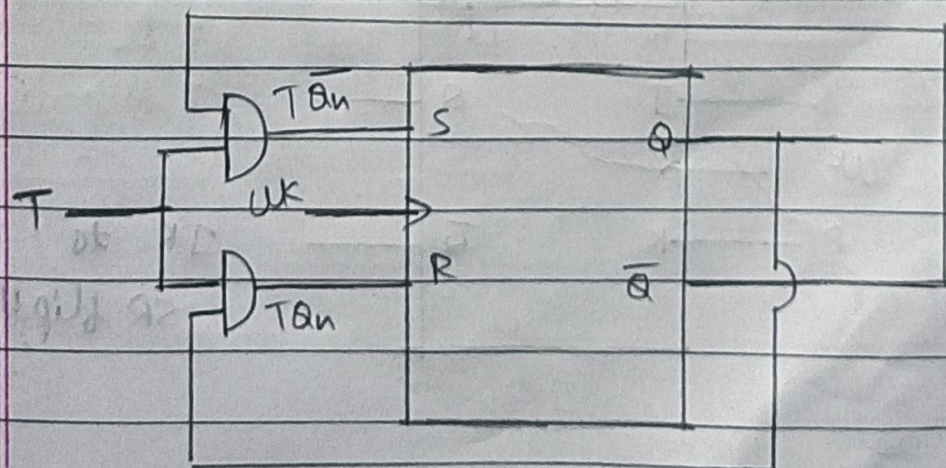
→ K-Map S

S_n	Q_n	0	1
0			X
1		1	

$$S = T \bar{Q}_n$$

R_n	Q_n	0	1
0		X	
1			1

$$R = T Q_n$$



2. JK flip flop to other flipflops

① JK to SR flipflop

→ The required flip flop is SR flip flop

→

S	R	Q_n	Q_{n+1}	J	K
0	0	0	0	0	X
0	0	1	1	X	0
0	1	0	0	0	X
0	1	1	0	X	1
1	0	0	1	1	X
1	0	1	1	X	0
1	1	0	X	X	X
1	1	1	X	X	X

→ K-Maps

J's

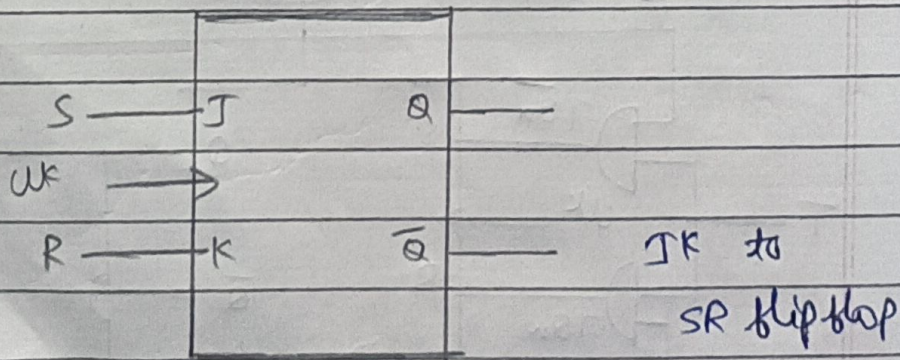
S \ R	0	1	1	0
0	0	X	X	
1	1	X	X	X

$J = S$

K's

S \ R	0	1	1	0
0	X		1	X
1	X		X	X

$K = R$



② JK to D flip flop

→ the required flip flop is D flip flop

→

	D	Q_n	Q_{n+1}	J	K
	0	0	0	0	X
	0	1	0	X	X
	1	0	1	X	X
	1	1	1	X	0

→ K-Maps

J =

D	Q_n
0	X
1	X

J = 1

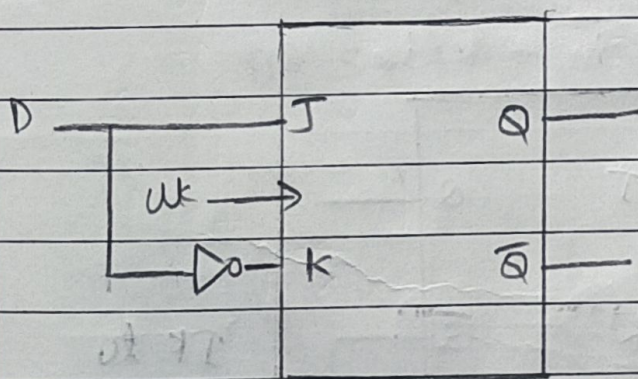
$$J = D$$

K =

D	Q_n
0	X
1	X

K = 1

$$K = \bar{D}$$



JK as

D flip flop

③ JK to T flip flop

→ the required flip flop is T flip flop

→

T	Q_n	Q_{n+1}	J	K
0	0	0	0	X
0	1	1	X	0
1	0	1	1	X
1	1	0	X	1

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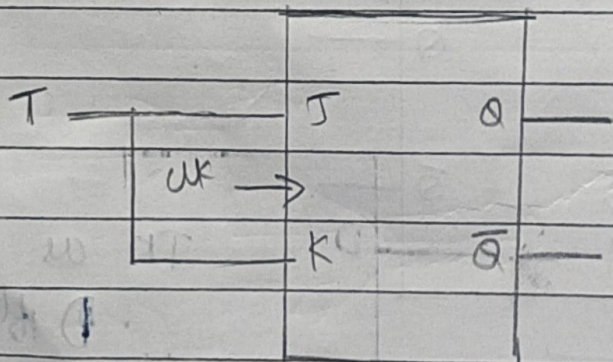
→ K-Maps

J \ Q	0	1
0		X
1	1	X

$J = T$

K \ Q	0	1
0	X	
1	X	1

$K = T$



JK to
T flip flop

3. D flip flop to other flipflops

① D to SR flip flop

→ the required flipflop is SR flipflop

→

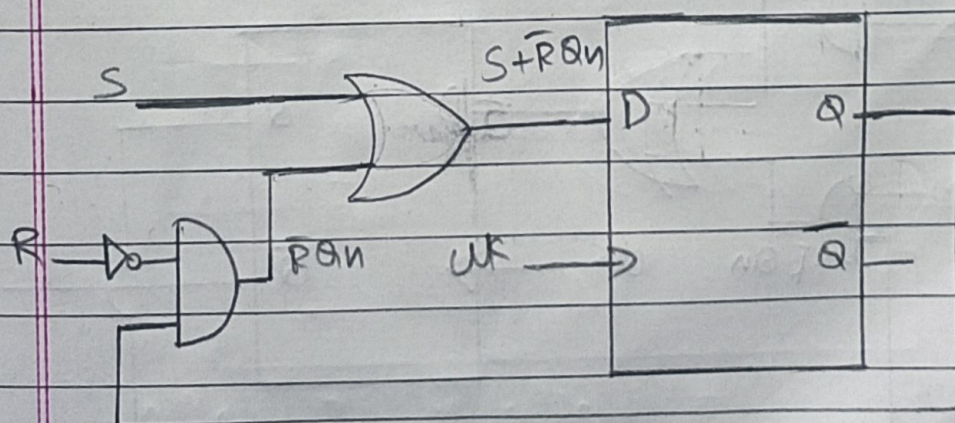
S	R	Q_n	Q_{n+1}	D
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	0	0
1	0	0	1	1
1	0	1	1	1
1	1	0	X	X
1	1	1	X	X

→ K-Map

D = $S + \bar{R}Q_n$

D	S \ R	00	01	11	10
0			1		
1		1	1	X	X

$$D = S + \bar{R}Q_n$$



D to SR flip flop

② D to JK flipflop

→ the required flipflop is JK flipflop

→

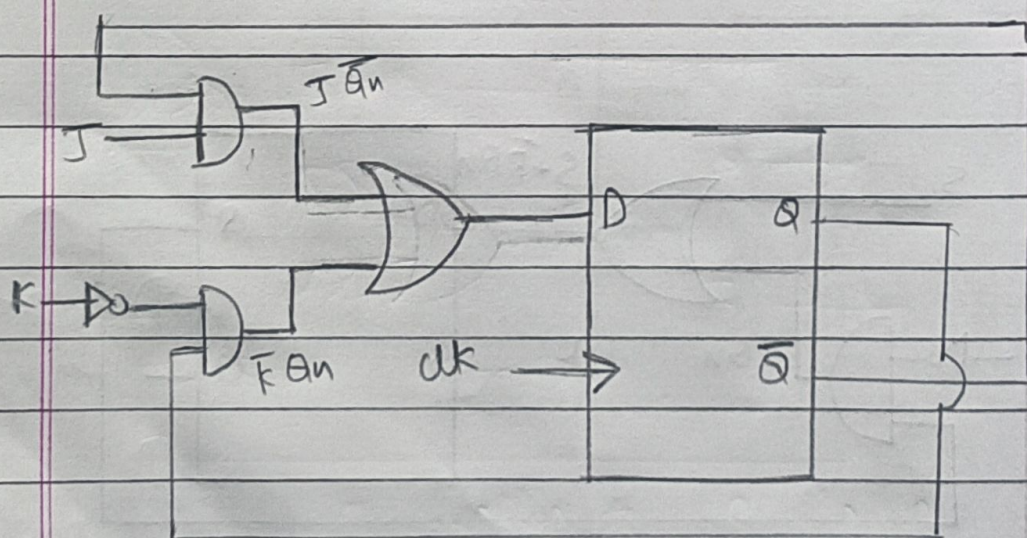
J	K	Q_n	Q_{n+1}	D
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	0	0
1	0	0	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

→ K-Map

D:-

$\bar{J}$	$\bar{K}Q_n$	00	01	11	10
0			1		
1		1	1		1

$$D = J\bar{Q}_n + \bar{K}Q_n$$



③ D to T flip flop

→ the required flip flop is T flip flop

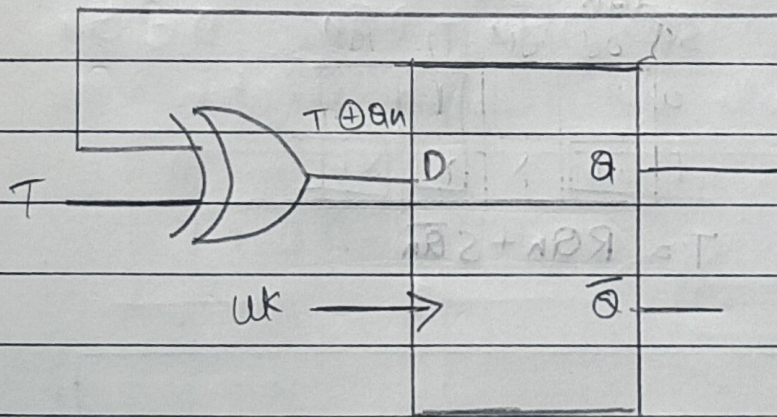
→

T	Q_n	Q_{n+1}	D
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	0

→ K-Map

D = $T\bar{Q}_n + \bar{T}Q_n$
 $D = T \oplus Q_n$

T \ Q_n	0	1
0	0	1
1	1	0



D to T flip flop

4. T flipflop to other flipflops

① T to SR flipflop

→ the required flipflop is SR flipflop

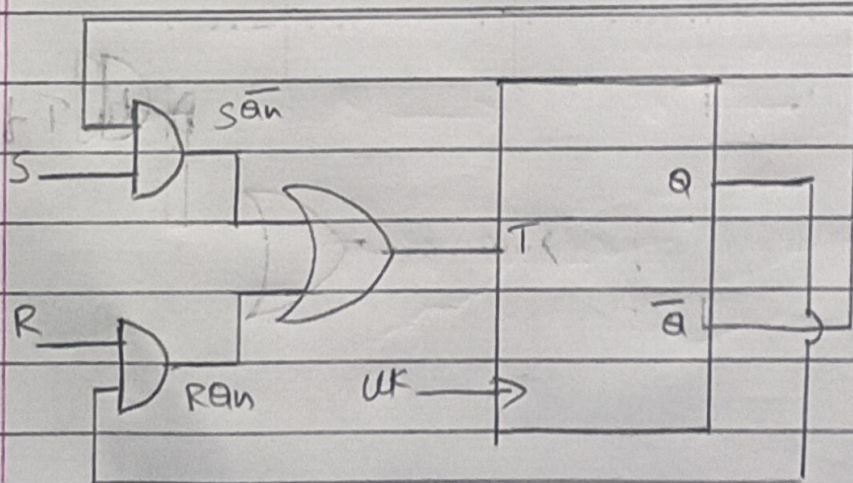
→

S	R	A_n	A_{n+1}	T
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	1
1	0	1	1	0
1	1	0	X	X
1	1	1	X	X

→ K-Map

T	S/R	00	01	11	10
0		1		1	
1		1		X	X

$$T = RA_n + S\bar{A}_n$$



T flipflop to SR flipflop

② T to JK flip flop

→ the required flip flop is JK flip flop.

→

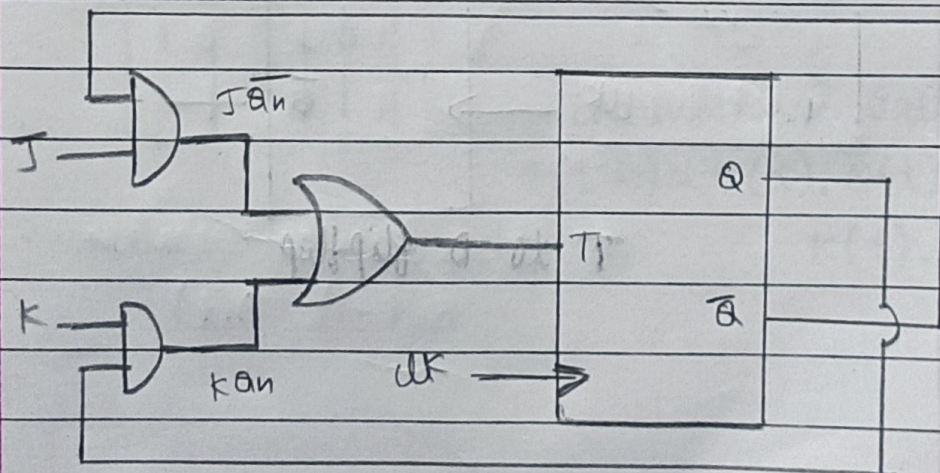
J	K	Q_n	Q_{n+1}	T
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	1
1	0	1	1	0
1	1	0	1	1
1	1	1	0	1

→ K-Map

T =

J \ Q_n	00	01	11	10
0	0	0	1	0
1	1	1	1	1

$$T = J\bar{Q}_n + KQ_n$$



③ T to d flip flop

→ The required flip flop is D flip flop

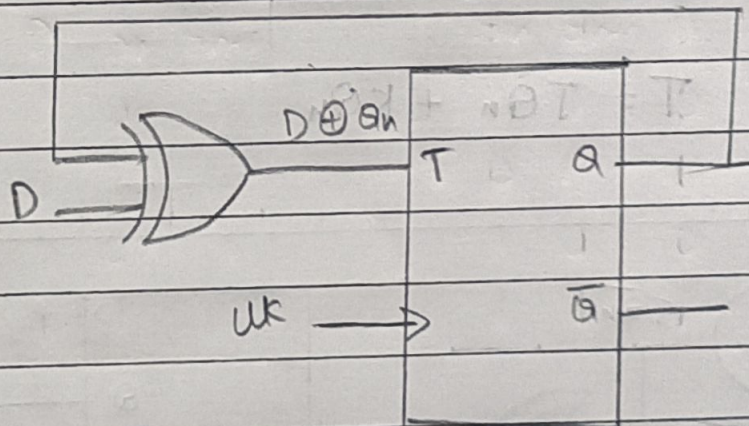
→

D	Q_n	Q_{n+1}	T
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

→ K-Map

T \ D	Q_n	1
0	1	0
1	1	0

$$T = D\bar{Q}_n + \bar{D}Q_n = D \oplus Q_n$$



T to D flip flop

* TRUTH TABLE Vs EXCITATION TABLE

→ Truth table gives the information about the output for a specific combination of input signals.

→ Excitation table gives the information about the set of inputs to be applied for getting a specific output. Excitation table will be very helpful for designing sequential circuits.

① JK Flipflop:-

→ Truth Table:-

J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

→ Excitation Table:-

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Characteristic equation:-

$$Q(t+1) = J(t) \cdot \bar{Q}(t) + K(t) \cdot Q(t)$$

② SR flip-flop

→ Truth Table

S	R	Q _n	Q _{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

→ Excitation Table

Q _n	Q _{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

characteristic equation

$$Q(t+1) = Q(t) \cdot R(t) + S(t)$$

③ D Flipflop

→ Truth Table

X	D	Q _n	Q _{n+1}
1	0	0	0
0	0	1	0
	1	0	1
	1	1	1

→ Excitation Table

Q _n	Q _{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

characteristic equation:-

$$Q(t+1) = D(t)$$

④ T- flipflop

→ Truth Table

T	Q _n	Q _{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

→ Excitation Table

Q _n	Q _{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

characteristic equation:-

$$Q(t+1) = Q(t) \oplus T(t)$$